

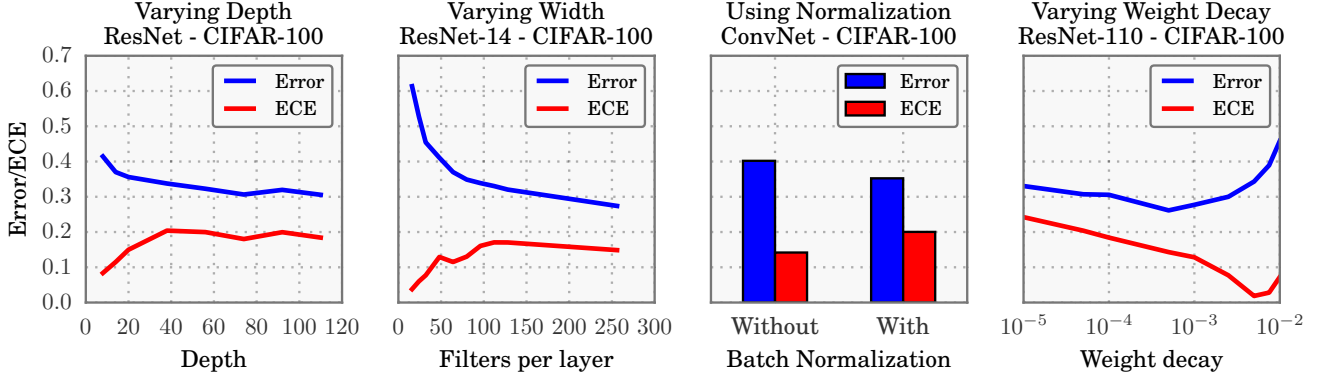


Figure 2. The effect of network depth (far left), width (middle left), Batch Normalization (middle right), and weight decay (far right) on miscalibration, as measured by ECE (lower is better).

taking a weighted average of the bins’ accuracy/confidence difference. More precisely,

$$\text{ECE} = \sum_{m=1}^M \frac{|B_m|}{n} \left| \text{acc}(B_m) - \text{conf}(B_m) \right|, \quad (3)$$

where n is the number of samples. The difference between acc and conf for a given bin represents the calibration *gap* (red bars in reliability diagrams – e.g. Figure 1). We use ECE as the primary empirical metric to measure calibration. See Section S1 for more analysis of this metric.

Maximum Calibration Error (MCE). In high-risk applications where reliable confidence measures are absolutely necessary, we may wish to minimize the worst-case deviation between confidence and accuracy:

$$\max_{p \in [0,1]} \left| \mathbb{P} \left(\hat{Y} = Y \mid \hat{P} = p \right) - p \right|. \quad (4)$$

The Maximum Calibration Error (Naeini et al., 2015) – or MCE – estimates this deviation. Similarly to ECE, this approximation involves binning:

$$\text{MCE} = \max_{m \in \{1, \dots, M\}} |\text{acc}(B_m) - \text{conf}(B_m)|. \quad (5)$$

We can visualize MCE and ECE on reliability diagrams. MCE is the largest calibration gap (red bars) across all bins, whereas ECE is a weighted average of all gaps. For perfectly calibrated classifiers, MCE and ECE both equal 0.

Negative log likelihood is a standard measure of a probabilistic model’s quality (Friedman et al., 2001). It is also referred to as the cross entropy loss in the context of deep learning (Bengio et al., 2015). Given a probabilistic model $\hat{\pi}(Y|X)$ and n samples, NLL is defined as:

$$\mathcal{L} = - \sum_{i=1}^n \log(\hat{\pi}(y_i | \mathbf{x}_i)) \quad (6)$$

It is a standard result (Friedman et al., 2001) that, in expectation, NLL is minimized if and only if $\hat{\pi}(Y|X)$ recovers the ground truth conditional distribution $\pi(Y|X)$.

3. Observing Miscalibration

The architecture and training procedures of neural networks have rapidly evolved in recent years. In this section we identify some recent changes that are responsible for the miscalibration phenomenon observed in Figure 1. Though we cannot claim causality, we find that increased model capacity and lack of regularization are closely related to model miscalibration.

Model capacity. The model capacity of neural networks has increased at a dramatic pace over the past few years. It is now common to see networks with hundreds, if not thousands of layers (He et al., 2016; Huang et al., 2016) and hundreds of convolutional filters per layer (Zagoruyko & Komodakis, 2016). Recent work shows that very deep or wide models are able to generalize better than smaller ones, while exhibiting the capacity to easily fit the training set (Zhang et al., 2017).

Although increasing depth and width may reduce classification error, we observe that these increases negatively affect model calibration. Figure 2 displays error and ECE as a function of depth and width on a ResNet trained on CIFAR-100. The far left figure varies depth for a network with 64 convolutional filters per layer, while the middle left figure fixes the depth at 14 layers and varies the number of convolutional filters per layer. Though even the smallest models in the graph exhibit some degree of miscalibration, the ECE metric grows substantially with model capacity. During training, after the model is able to correctly classify (almost) all training samples, NLL can be further minimized by increasing the confidence of predictions. Increased model capacity will lower training NLL, and thus the model will be more (over)confident on average.

Batch Normalization (Ioffe & Szegedy, 2015) improves the optimization of neural networks by minimizing distribution shifts in activations within the neural network’s hid-